TITLE PAGE

- **Problem Statement ID –** Q4A
- **Problem Statement Title-** AI-Powered Behavioral Anomaly Detection for Cybersecurity
- **Theme-** Cybersecurity
- **PS Category-** Software
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IDEA TITLE

SENTINELAI — Explainable Behavioural Anomaly Detection

- Learns normal access behaviour for users, service accounts and edge devices using login time, location, IP, device, resource and command-sequence patterns.
- Detects credential misuse, brute force, impossible travel, lateral movement, device spoofing and low-and-slow exfiltration; produces a 0–100 risk score.
- Provides analyst-ready explanations for every alert and adapts safely to legitimate behaviour changes and new users/devices.

TECHNICAL APPROACH

- **Technologies:** Python, pandas, NumPy, scikit-learn, Streamlit, Plotly, Matplotlib and joblib; CSV datasets and saved ML models.

- **Implementation flow:**
  Synthetic access logs → Per-entity behavioural baseline + temporal/sequence features → Isolation Forest + balanced Random Forest → Risk score, attack category and explanation → SOC dashboard, alerts and evaluation report

FEASIBILITY AND VIABILITY

- **Feasible prototype:** Built with open-source Python tools and tested on **4,182** synthetic access events; produces real-time-style risk scores, ranked alerts and explanations.

- **Key risks:** Synthetic data may not represent all real attacks; false positives can occur for new users, changing work patterns and incomplete logs.

- **Mitigation strategy:** Use population baselines for cold starts, adaptive behaviour profiles for drift, analyst feedback for tuning, and periodic retraining on securely collected production logs.

ARTIFACTS

- **Source code:** Synthetic data generator, feature-engineering pipeline, anomaly detector, evaluation module and Streamlit dashboard.
- **Model artefacts:** Saved Isolation Forest and Random Forest models; per-entity baseline profiles and scored access logs.
- **Evidence:** Automated test results, evaluation report, ROC/precision–recall/confusion-matrix charts and dashboard screenshots.

RESEARCH AND REFERENCES

scikit-learn — Isolation Forest:
https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html

scikit-learn — Random Forest Classifier:
https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

MITRE ATT&CK — Brute Force / Credential Stuffing:
https://attack.mitre.org/techniques/T1110/

Streamlit Documentation — Analyst Dashboard:
https://docs.streamlit.io/